Xin (Allen) Wang

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Education

Yale University — Aug 2025 – May 2030
Ph.D. Computational Biology and Bioinformatics — *New Haven, CT*

Vanderbilt University — Aug 2023 – May 2025
M.S. Computer Science, GPA: 4.0/4.0 — *Nashville, TN*

- Thesis Track: *Data-driven Machine Learning for Small Molecule Drug Discovery*

University of California, Santa Barbara — Aug 2019 – Jun 2023
B.S. Statistics and Data Science, Overall GPA: 3.88/4.0, Major GPA: 3.92/4.0 — *Goleta, CA*

Research Experience

Yale Biomedical Informatics & Data Science — Sep 2025 – Present
Ph.D. Student — *Yale University*
Advisor: Dr. Qingyu Chen
Research Interests: AI Scientists and RSI for Scientific Discovery (SciAgentArena)

Center for AI in Protein Dynamics — May 2025 – Present
Researcher — *Vanderbilt University*
Advisor: Dr. Carlos Oliver
Research Interests: Protein Substructure Discovery (BioBlobs), RNA Inverse Design (KnitNet)

Network and Data Science (NDS) Lab — Aug 2023 – May 2025
Graduate Research Assistant — *Vanderbilt University*
Advisors: Dr. Tyler Derr, Dr. Jens Meiler
Research Interests: Data-centric AI for Drug Discovery (ScaffAug, WelQrate)

Caves Lab/Data Science Capstone — Nov 2022 – Jun 2023
Undergraduate Research Assistant — *UC Santa Barbara*
Advisor: Dr. Eleanor Caves
Research Interests: Computer Vision, Biological Image Analysis (BeeODA)

Publications

†Equal contribution

1. **Xin Wang**, Kaiwen Shi, and Carlos Oliver. "BioBlobs: Unsupervised Discovery of Functional Substructures." In Advances in Neural Information Processing Systems (**NeurIPS '26**).

2. **Xin Wang**, Yu Wang, Yunchao Liu, Jens Meiler, and Tyler Derr. "Scaffold-Aware Generative Augmentation and Reranking for Enhanced Virtual Screening." In Proceedings of the SIAM International Conference on Data Mining (**SDM '26**).

3. Tianyu Liu†, **Xin Wang**†, Antonia Panescu†, et al. "Benchmarking AI Agents for Addressing Scientific Challenges Across Scales." Submitted to *Nature*, 2026. arXiv:2606.12736.

4. Qingyu Chen, **Xin Wang**, Rong Zhou, et al. "AI Scientists in Health: Trustworthy Agentic AI for Scientific Inquiry." Submitted to *Nature Biomedical Engineering*, 2026. SSRN:6889798.

5. Antonia Panescu, **Xin Wang**, Haoran Xu, Qingyu Chen, Carlos Oliver, and Rex Ying. "KnitNet: Learning to Search at Inference Time for Exact RNA Inverse Design." Under submission, 2026.

6. Yunchao Liu†, Ha Dong†, **Xin Wang**†, Rocco Moretti, Yu Wang, Zhaoqian Su, Jiawei Gu, Bobby Bodenheimer, Charles David Weaver, Jens Meiler, and Tyler Derr. "WelQrate: Defining the Gold Standard in Small Molecule Drug Discovery Benchmarking." In Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track (**NeurIPS '24**), 53222–53236.

7. Yu Wang, Nedim Lipka, Ruiyi Zhang, Alexa Siu, Yuying Zhao, Bo Ni, **Xin Wang**, Ryan Rossi, and Tyler

Derr. "Topology-aware Retrieval Augmentation for Text Generation." In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (**CIKM '24**), 2442–2452.

8. Clayton W. Kosonocky, Alex M. Abel, et al. "Validation and Analysis of 12,000 AI-driven CAR-T Designs in the Bits to Binders Competitions." *bioRxiv*, 2026. (Consortium author, Bits to Binders Competitors)

Talks

AI Scientist Summer Workshop Aug 2026
Contributed Talk: Benchmarking AI Agents for Addressing Scientific Challenges Across Scales *Cambridge, MA*

Work Experience

Summer Researcher Jun 2024 – Aug 2024
Vanderbilt University, Computer Science Department *Nashville, TN*

- Analyzed ego-subgraph topology of anomalous nodes across benchmark datasets and improved graph anomaly detection by augmenting local topology with a diffusion model.
- Contributed to an Amazon Research Awards proposal, *Generative AI for Graph Anomaly Detection.*

Undergraduate Teaching Assistant Sep 2022 – Jun 2023
UC Santa Barbara, Dept. of Statistics & Applied Probability *Goleta, CA*

- Supported courses in Statistical Machine Learning, Regression Analysis, and Big Data Analytics.
- Led lab sections and office hours, helping students with coursework and programming.

Algorithm Engineer Intern Jun 2022 – Aug 2022
Ping An Technology, Intelligent City Group *Shenzhen, China*

- Mined and analyzed monthly city-level macroeconomic data and trained LSTM-based forecasting models.
- Curated and preprocessed the policy database powering a policy recommendation system.

Algorithm Intern Jun 2021 – Aug 2021
Bosera Funds, Index and Quantitative Department *Shenzhen, China*

- Collected and modeled daily stock transaction data for quantitative analysis.

Technical Skills

Methods: Protein & RNA Language Models, Diffusion Language Models, GNNs, LLM Fine-tuning, Agent Evaluation
Programming: Python, C/C++, SQL, R
ML Frameworks: PyTorch (DDP), PyTorch Geometric, Hugging Face Transformers, PEFT/LoRA, DeepSpeed, vLLM
AI for Biology: ESM/ESMFold, AlphaFold, RFdiffusion, ProteinMPNN, Chai-1, ViennaRNA, PyMOL
Agent Evaluation & HPC: Harbor, Terminal-Bench, Slurm, Docker, Apptainer, Git, Linux, Weights & Biases

Honors & Awards

Vanderbilt Graduate School Travel Grant ($1,000) Nov 2024
Vanderbilt Graduate Fellowship ($20,000) Sep 2023
College Honors: UCSB Arts and Sciences Jun 2023
People's Choice Award: UCSB Math Directed Reading Program (DRP) Jun 2022

Services

Reviewer: ICLR 2026, ICLR 2027
Subreviewer: NeurIPS 2024, CIKM 2024, SDM 2024, WWW 2024, ECML-PKDD 2024
PC Member: GTA^3-2024, WSDM 2024-MLoG, MLoG-GenAI (KDD 2025)
Board Member: Vanderbilt University Riichi Mahjong League
Mentor: UCSB Mentorship Program
Audit Director: Shenzhen Middle School Student Activity Association